ENEL 585 – Introduction to Power Electronics

Fall 2026 · Quiz 2 · Tuesday, October 13, 2026 · Worked solutions

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This is the quiz as written, with a worked solution in the box under each part. The quiz has **3 questions** worth **11 marks**, plus two **bonus** parts worth 1 mark each; it is graded **out of 11**.

Conditions that applied to every part

- Unless a part says otherwise: all components and switches are ideal (no losses, no dead time) and the circuit is in periodic steady state. The fundamental frequency is $f_1 = 60$ Hz ($\omega_1 = 377$ rad/s).
- **Show your working.** A correct final number with no working shown earned at most half the marks for that part. One line is enough, for example the table entry you read, the impedance you evaluated or the equation you set up.

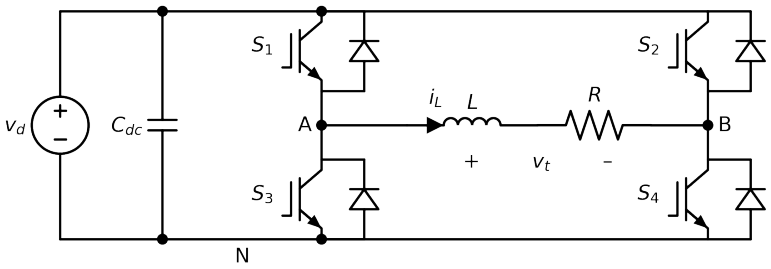
Notation (as in the lectures): v_d dc-link voltage; v_t bridge output voltage ($v_t = v_{AN} - v_{BN}$); a hat denotes a **peak** value and phasors use peak magnitudes, e.g. $\underline{V}_t = \hat{V}_t \angle\theta$; subscript h is the harmonic order ($h = 1$ is the fundamental, at f_1; harmonic h is at hf_1); m_i modulation index; $m_f = f_{sw}/f_1$.

The ideas used throughout

- **Each harmonic separately:** the load is linear, so each harmonic of v_t drives its own current through the load impedance **at that harmonic's frequency**, $Z_h = R + jh\omega_1 L$.
- **Phasors with peak magnitudes:** KVL in phasor form, and average power $P = \frac{1}{2}\hat{V}\hat{I}\cos\varphi$.
- **Power balance across the bridge:** the bridge is lossless, so the average dc power equals the average ac power. In a single-phase circuit the ac power also pulsates at $2\omega_1$, and the dc-link capacitor buffers that pulsation.
- **Choosing a switch:** when the switch is on, the sign of i_{sw} it must conduct; when it is off, the sign of v_{sw} it must block. Together they fix the quadrants, and the device must cover them all.

Question 1 – The Lab 2 bridge with sinusoidal PWM (3 marks)

The Lab 2 inverter is supplied from $v_d = 55$ V and feeds the series load $L = 5$ mH, $R = 12.5\ \Omega$. It is operated with **bipolar** sinusoidal PWM at $m_i = 0.6$ and a carrier frequency $f_{sw} = 4.98$ kHz, so $m_f = 83$.



The table below gives the harmonics of v_t for bipolar sinusoidal PWM as $c_h = \hat{V}_{t,h}/v_d$. A row marked $\pm$ gives the value of each of its two components.

h	$m_i = 0.2$	0.4	0.6	0.8	1.0
1	0.200	0.400	0.600	0.800	1.000
m_f	1.242	1.150	1.006	0.818	0.601
$m_f \pm 2$	0.016	0.061	0.131	0.220	0.318
$2m_f \pm 1$	0.190	0.326	0.370	0.314	0.181
$2m_f \pm 3$	–	0.024	0.071	0.139	0.212

(a) [1] Determine the peak value of the fundamental component of the load current i_L.

Solution

Fundamental voltage (row $h = 1$, column $m_i = 0.6$): $\hat{V}_{t,1} = m_i v_d = 0.6 \times 55 = 33.0$ V.
Load impedance at 60 Hz: $\omega_1 L = 377 \times 5$ mH $= 1.885\ \Omega$, so

$$|Z_1| = \sqrt{R^2 + (\omega_1 L)^2} = \sqrt{12.5^2 + 1.885^2} = 12.64\ \Omega$$

$$\hat{I}_{L,1} = \frac{\hat{V}_{t,1}}{|Z_1|} = \frac{33.0\text{ V}}{12.64\ \Omega} = \mathbf{2.61\ A}\ \text{(peak)}$$

At 60 Hz the load is almost purely resistive: neglecting $\omega_1 L$ would give 2.64 A.

(b) [2] Apart from the fundamental, which harmonic component of v_t has the largest amplitude? Give its harmonic order and its frequency, and the peak value of the component of i_L at that frequency. Then estimate the THD of i_L, counting only this harmonic.

Solution

Which harmonic. In the $m_i = 0.6$ column, the largest entry after the fundamental is the m_f row, $c_h = 1.006$. So the harmonic of v_t with the largest amplitude is

$$h = m_f = \mathbf{83}, \qquad f = 83 \times 60\ \text{Hz} = \mathbf{4980\ Hz}\ \text{(the carrier frequency)}$$

with amplitude $\hat{V}_{t,83} = 1.006 \times 55 = 55.3$ V.
Its current. The impedance has to be evaluated **at 4980 Hz**, where the inductor's reactance is 83 times its 60 Hz value:

$$|Z_{83}| = \sqrt{12.5^2 + (83 \times 1.885)^2} = \sqrt{12.5^2 + 156.5^2} = 157\ \Omega$$

$$\hat{I}_{L,83} = \frac{55.3\ \text{V}}{157\ \Omega} = \mathbf{0.35\ A}$$

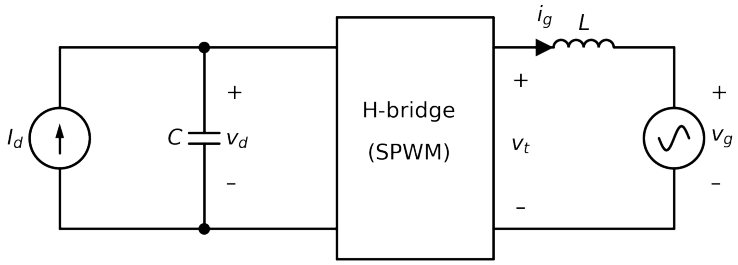
THD, counting only this harmonic:

$$\text{THD}_i \approx \frac{\hat{I}_{L,83}}{\hat{I}_{L,1}} = \frac{0.35}{2.61} = \mathbf{13.5}\ \%$$

The ratio is the same whether both currents are peak or both are rms; it must not mix the two. Including every row of the table raises the THD only to about 14 %, so counting one harmonic is a fair estimate.
This harmonic **voltage** (55.3 V) is larger than the fundamental (33.0 V), yet its **current** is seven times smaller. The series inductor does the filtering: its impedance rises with frequency, which is why a high carrier frequency makes the load current nearly sinusoidal.
Pitfall: dividing 55.3 V by the 60 Hz impedance, 12.64 Ω. That gives 4.38 A, a harmonic current larger than the fundamental, which should be a warning sign.

Question 2 – A grid-connected inverter and its dc link (6 marks + 1 bonus)

An H-bridge with sinusoidal PWM is connected to the grid through $L = 5$ mH. The grid voltage is $v_g(t) = 170\cos(\omega_1 t)$ V. On the dc side, a source supplies a **constant** current I_d and the dc-link capacitor C holds the voltage near $v_d = 250$ V. Consider only the fundamental components of v_t and i_g (neglect all switching-frequency components).



(a) [3] The inverter is to deliver $P = 1.5$ kW to the grid with i_g in phase with v_g. Determine the phasor $\underline{V}_{t,1}$ (peak magnitude and angle relative to v_g) that the bridge must produce, and the modulation index m_i.

Solution

Take v_g as the reference: $\underline{V}_g = 170\angle 0°$ V.

Grid current. With i_g in phase with v_g, $P = \frac{1}{2}\hat{V}_g\hat{I}_g$ (peak values), so

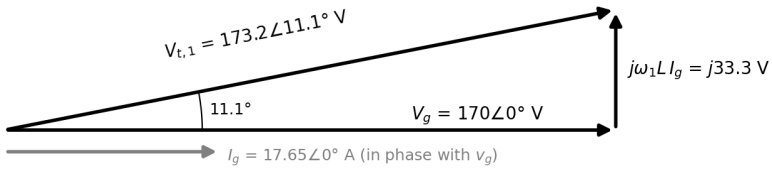
$$\hat{I}_g = \frac{2P}{\hat{V}_g} = \frac{2 \times 1500}{170} = 17.65 \text{ A}, \qquad \underline{I}_g = 17.65\angle 0° \text{ A}$$

KVL around the output loop, with $\omega_1 L = 1.885\ \Omega$:

$$\underline{V}_{t,1} = \underline{V}_g + j\omega_1 L\,\underline{I}_g = 170 + j(1.885)(17.65) = 170 + j33.3 \text{ V}$$

$$\underline{V}_{t,1} = \mathbf{173.2\angle 11.1°\ V} \text{ (peak; } v_t \text{ leads } v_g\text{)}$$

The inductor voltage leads its current by 90°, so the 33.3 V drop is added **at right angles** to $\underline{V}_g$, not to its magnitude. The bridge needs only 3 V more than the grid, but it must lead the grid by 11°: that angle is what sends power into the grid.



Modulation index (sinusoidal PWM, linear range, $\hat{V}_{t,1} = m_i v_d$):

$$m_i = \frac{\hat{V}_{t,1}}{v_d} = \frac{173.2}{250} = \mathbf{0.693}$$

In part (b) and in the bonus part (c), neglect the reactive power absorbed by L: take the instantaneous power at the bridge output to be the same as at the grid terminals.

(b) [3] For the operating point of part (a), determine I_d, and the smallest C that keeps the peak-to-peak ripple of v_d within 10 V.

Solution

dc current. The bridge is lossless and the capacitor carries no average current in steady state, so the average dc power equals the average ac power:

$$v_d I_d = P \quad \Rightarrow \quad I_d = \frac{1500\ \text{W}}{250\ \text{V}} = \mathbf{6.0\ A}$$

What the capacitor does. With i_g in phase with v_g, the instantaneous power is

$$p(t) = \hat{V}_g \hat{I}_g \cos^2(\omega_1 t) = \underbrace{P}_{\text{average}} + \underbrace{P\cos(2\omega_1 t)}_{\text{twice line frequency}}$$

The source supplies only the constant part, $v_d I_d = P$. The capacitor supplies the whole $2\omega_1$ term: it discharges while $p > P$ and charges while $p < P$.
Sizing (as in the lectures). Over one positive lobe of the $2\omega_1$ term the capacitor gives up

$$\Delta E = \int_0^{\pi/2\omega_1} P\sin(2\omega_1 t)\, dt = \frac{P}{\omega_1}$$

and from the stored energy $\frac{1}{2}Cv_d^2$, that energy change moves v_d through its full peak-to-peak ripple: $\Delta E = C\, v_d\, \Delta v_d$. With $\Delta v_d = 10$ V peak-to-peak:

$$C \geq \frac{P/\omega_1}{v_d\, \Delta v_d} = \frac{1500/377}{250 \times 10} = \frac{3.98\ \text{J}}{2500\ \text{V}^2} = \mathbf{1.59\ mF}$$

(c) [**bonus**, 1] The controller is changed so that i_g keeps the same amplitude as in part (a) but **leads** v_g by 90°. The capacitor is the one chosen in part (b). In the new steady state, what is I_d, and what is the peak-to-peak ripple of v_d?

Solution

dc current. With i_g at 90° to v_g, the average power is $P = \frac{1}{2}\hat{V}_g\hat{I}_g\cos 90° = 0$, so $I_d = \mathbf{0}$.
Ripple. The power pulsation does **not** vanish. With $i_g = -17.65\sin(\omega_1 t)$:

$$p(t) = 170\cos(\omega_1 t)\cdot(-17.65\sin(\omega_1 t)) = -\tfrac{1}{2}(170)(17.65)\sin(2\omega_1 t) = -1500\sin(2\omega_1 t)\ \text{W}$$

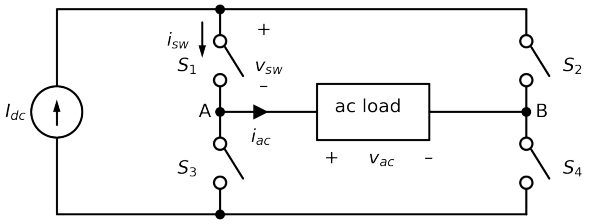
Its amplitude is the apparent power $S = \frac{1}{2}\hat{V}_g\hat{I}_g = 1500$ VA, the same as the amplitude of the $2\omega_1$ term in part (b). In general the $2\omega_1$ term has amplitude S, not P. The energy swing is again S/ω_1, so with the same capacitor the ripple is unchanged:

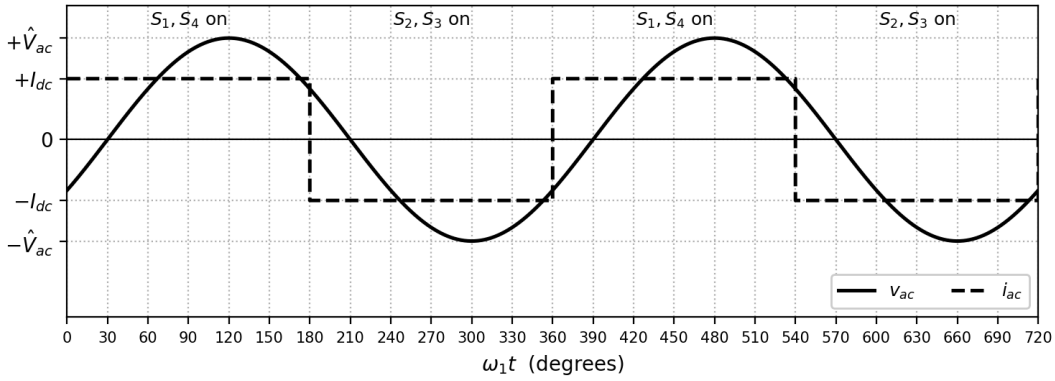
$$\Delta v_d = \frac{S/\omega_1}{C\,v_d} = \frac{1500/377}{1.59\ \text{mF}\times 250\ \text{V}} = \mathbf{10\ V}\ \text{peak-to-peak}$$

The capacitor is still needed even though the source supplies nothing, because it exchanges energy with the grid twice per line cycle.
Pitfall: "no real power, so no ripple". The average of $p(t)$ is zero, but $p(t)$ itself is not.

Question 3 – Choosing a switch (2 marks + 1 bonus)

The bridge below is fed from a dc **current** source $I_{dc} > 0$ instead of a voltage source. The diagonal pairs are switched at 60 Hz as marked on the waveforms, so i_{ac} is a square wave of amplitude I_{dc}. The ac load holds v_{ac} sinusoidal, and v_{ac} lags the fundamental of i_{ac} by 30°.





With v_{sw} and i_{sw} defined as shown for S_1:

(a) [2] Which sign(s) of i_{sw} must S_1 conduct when it is on, and which polarity (or polarities) of v_{sw} must it block when it is off? Show how you obtained each.

Solution

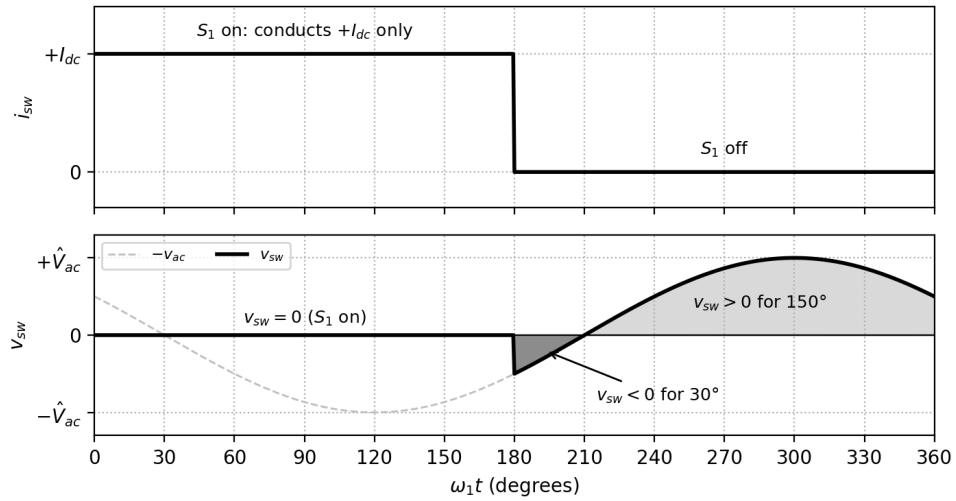
On state ($0° < \omega_1 t < 180°$, S_1 and S_4 on). S_1 carries the source current from the top rail into node A: $i_{sw} = +I_{dc}$. Since $I_{dc} > 0$, S_1 must conduct **positive current only**.
Off state ($180° < \omega_1 t < 360°$, S_2 and S_3 on). S_2 ties the top rail to B and S_3 ties the bottom rail to A, so the voltage across S_1 is

$$v_{sw} = v_{\text{top}} - v_A = v_B - v_A = -v_{ac}$$

Read v_{ac} from the waveform over that half period. Because v_{ac} lags by 30°, it does not change sign at 180° with the current; it crosses zero at 210°:

- 180° to 210°: $v_{ac} > 0$, so $v_{sw} < 0$ (for 30°);
- 210° to 360°: $v_{ac} < 0$, so $v_{sw} > 0$ (for 150°).

S_1 must block **both polarities**.



Pitfall: treating v_{ac} as if it were in phase with i_{ac}. v_{sw} would then be positive for the whole off-interval, and the 30° of negative voltage, which decides part (b), would be missed.

(b) [**bonus**, 1] Choose a device, or a combination of devices, from those covered in the lectures that can realize S_1. Say whether an IGBT with an antiparallel diode, as used in Lab 2, would be suitable, and why.

Solution

Required quadrants. Positive current, both voltage polarities: quadrants I and II.
Device. From the lectures' operating-region plots, this is the **thyristor** (blocks both polarities, conducts positive current). A thyristor cannot be turned off from its gate, but here the circuit does it: when S_2 and S_3 take over at 180°, v_{sw} is negative for the next 30°, which takes the current to zero and reverse-biases S_1 while it recovers.
If S_1 must also be turned off from its gate, quadrants I and II can be built from an **IGBT in series with a diode** (the diode adds reverse blocking), or a reverse-blocking IGBT. A bidirectional switch also works, though it covers all four quadrants and so is more than is needed.
IGBT with antiparallel diode: not suitable. It covers quadrants I and IV, so it cannot block negative voltage. During 180° to 210°, when $v_{sw} < 0$, its diode would be forward-biased and conduct. With S_2 on, the diode of S_1 and S_2 would short-circuit the ac load between A and B.

End of worked solutions.